

Question Number	Answer	Mark
7(a)	The only correct answer is D 20 000:1 <i>A is not correct because 100 cm is 20 000 times larger than 0.05mm</i> <i>B is not correct because the ratio is the wrong way round</i> <i>C is not correct because 100 cm is 20 000 times larger than 0.05mm</i>	(1)

Question Number	Answer	Additional guidance	Mark
7(b)(i)	An answer that includes the following point: group of (similar) cells working together to perform a (specific) function (1)	ACCEPT group of cells performing {the same /a similar} function	(1)

Question Number	Answer	Additional guidance	Mark
7(b)(ii)	An answer that includes the following points: - from the phloem (of vine plant) / phloem is involved in translocation / phloem is involved in transport of {sucrose / organic solutes} (1) (because part) of the corpse lily plant has {grown into the phloem / grown into vascular bundle / made a (physical) connection to the phloem} (1) (corpse lily plant absorbs organic solutes / sucrose) by {diffusion / facilitated diffusion / active transport} (1)	ignore xylem ACCEPT has digested (part of) the phloem ignore connecting cytoplasm ignore {grow in / attach to} {tissues/ xylem} of vine plant ACCEPT through plasmodesmata do not accept osmosis ignore using ATP	(3)

Question Number	Answer	Additional guidance	Mark
7(c)(i)	An explanation that includes the following points: - gel is used to apply the extract to the wound / extract dissolves in the gel (1) - gel has no effect on the rate of wound healing (1) - therefore can be used as a comparison (to extract / current wound treatment) (1)	ACCEPT gel acts as a base / helps extract stay on wound / helps application process / {faster / slow-release} absorption of extracts into {skin / wound} ACCEPT gel acts a {placebo / control} ignore controlled variable ACCEPT to make all of the treatments look the same (to scientist) / idea of blind trial	(3)

Question Number	Answer	Additional guidance	Mark
7(c)(ii)	An answer that includes four of the following points: - both the extracts and the current wound treatment {reduced the mean healing time (compared with just gel) / were more effective than just using gel} (1) - increasing the concentration of the extract decreased the time (taken for the wound to heal) / {little / no significant} difference in the healing time of the two extracts (1) - the current wound treatment is the most effective treatment / extract is less effective than the current wound treatment (1) - significant difference as the times for different groups do not overlap (1) - small sample size reduces validity of {conclusions / data / experiment} (1)	ignore just descriptions of the results ACCEPT {gel (only) / group 1} {had the longest healing time / was least effective} ignore current wound treatment had lowest healing time unqualified ACCEPT significant difference as the {standard deviations/data} do not overlap ACCEPT relevant comment about {size / overlap} of standard deviations linked to {repeatability / validity / reliability} (of data / investigation) ACCEPT a larger sample size should be used / not enough data collected to make a (valid) conclusion	(4)